Describing the relationship between two numerical variables

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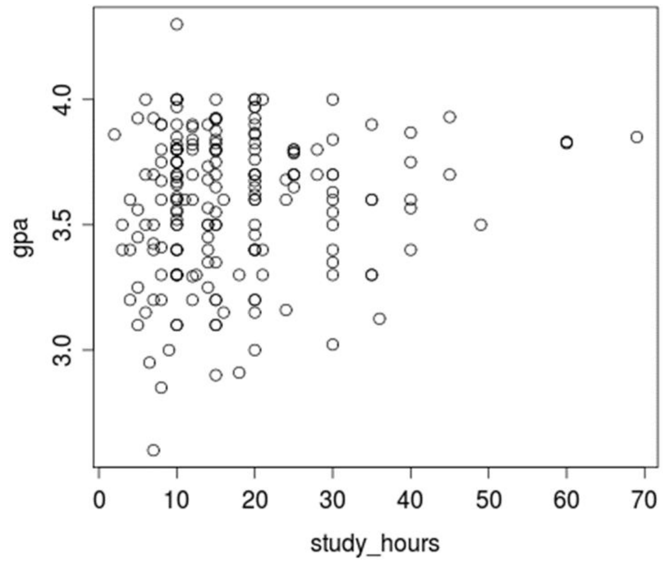
Slides include excerpts from Intro to Modern Statistics. Original slides modified by Jill Tysse, Hood College.

Modeling numerical variables

In this unit we will learn to quantify the relationship between two numerical variables, as well as modeling numerical response variables using a numerical or categorical explanatory variable.

Scatterplots



Describing scatterplots: Key Features

- **Shape,**
- **Strength,**
- **Direction**

of the association.

A ***scatterplot*** is the graph of the relationship between two quantitative variables.

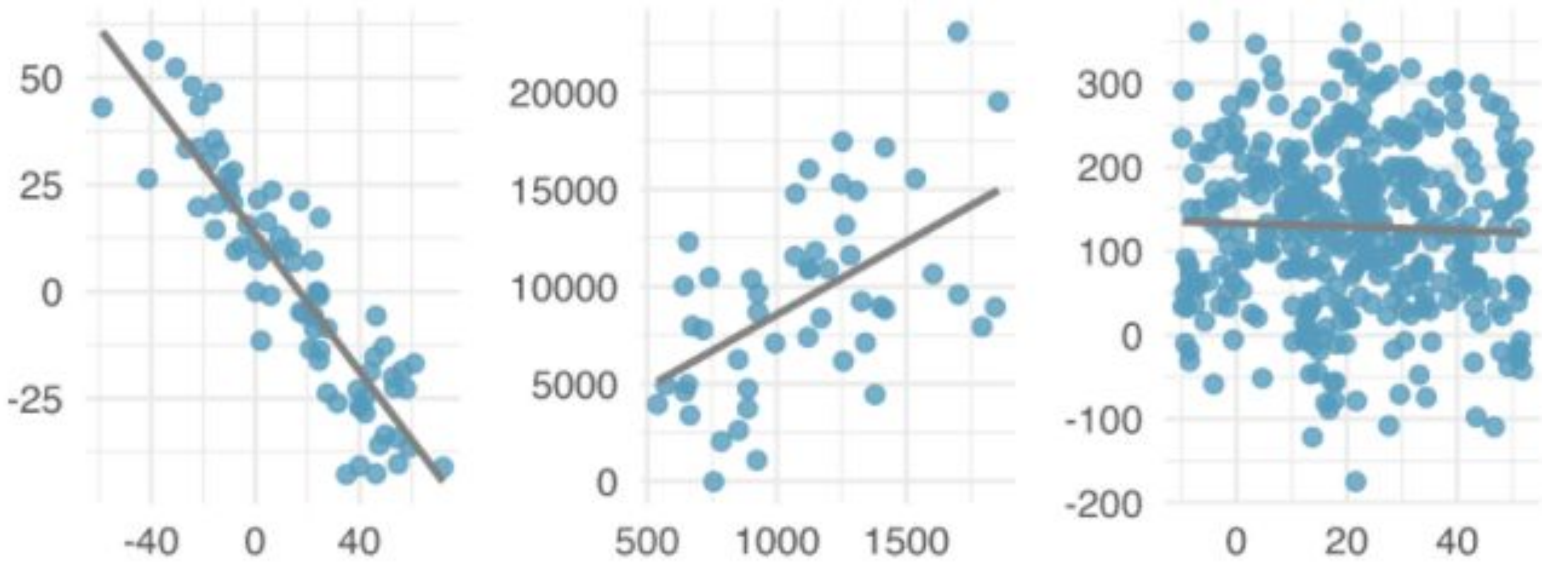
Shape, Strength, Direction

- **Shape:** Is the shape **linear**, or some other shape? *We will practice describing scatterplots of many shapes, but will focus on fitting curves to only linear graphs.*

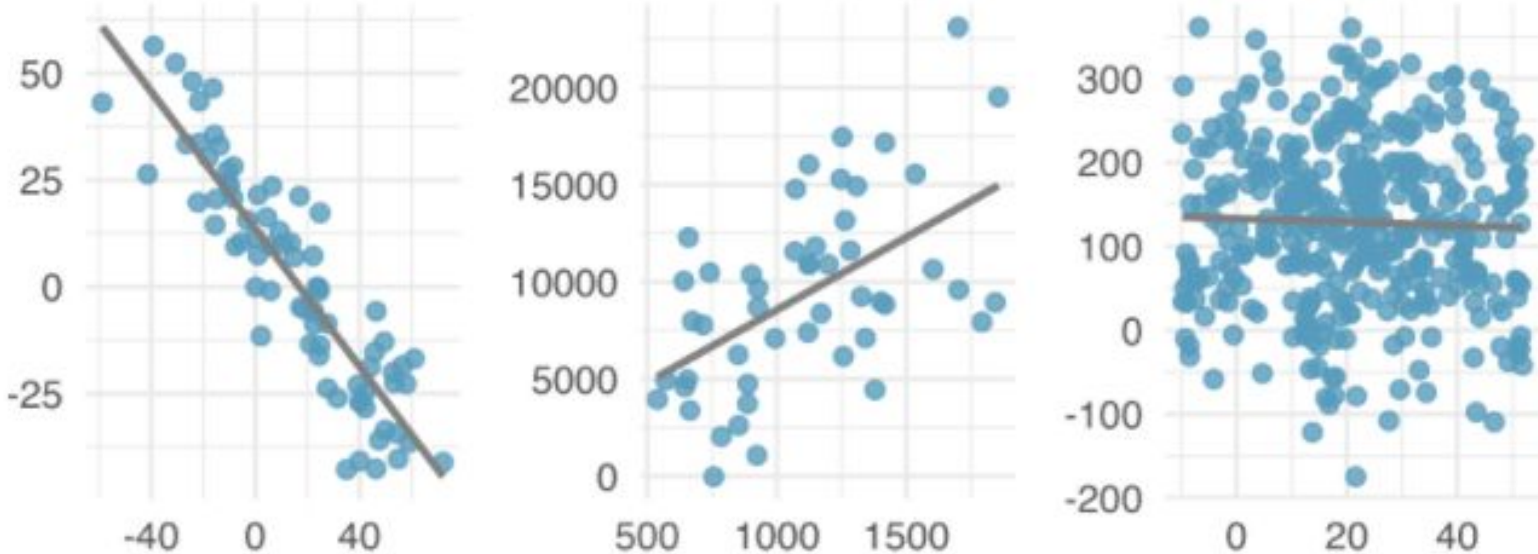
- **Strength:** Are the variables **strongly**, **moderately**, or **weakly** associated?

- **Direction:** Is the association **positive**, **negative**, or are the variables **not associated** at all?

Shape, strength, direction

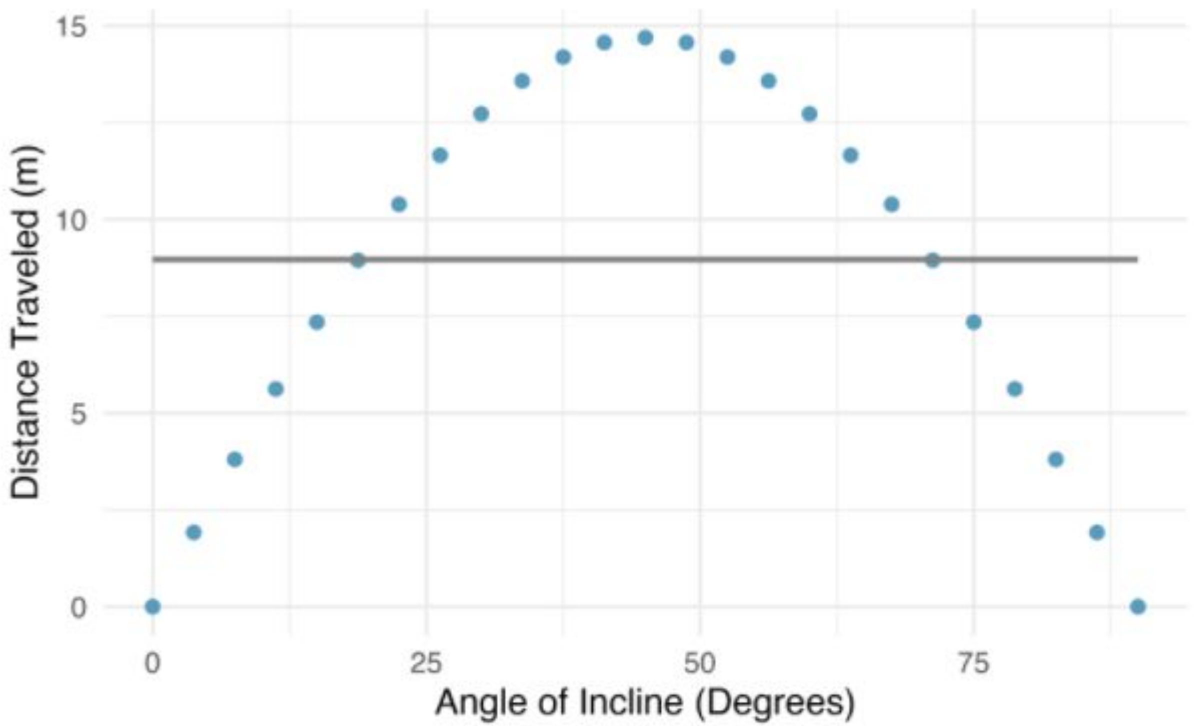


Shape, strength, direction



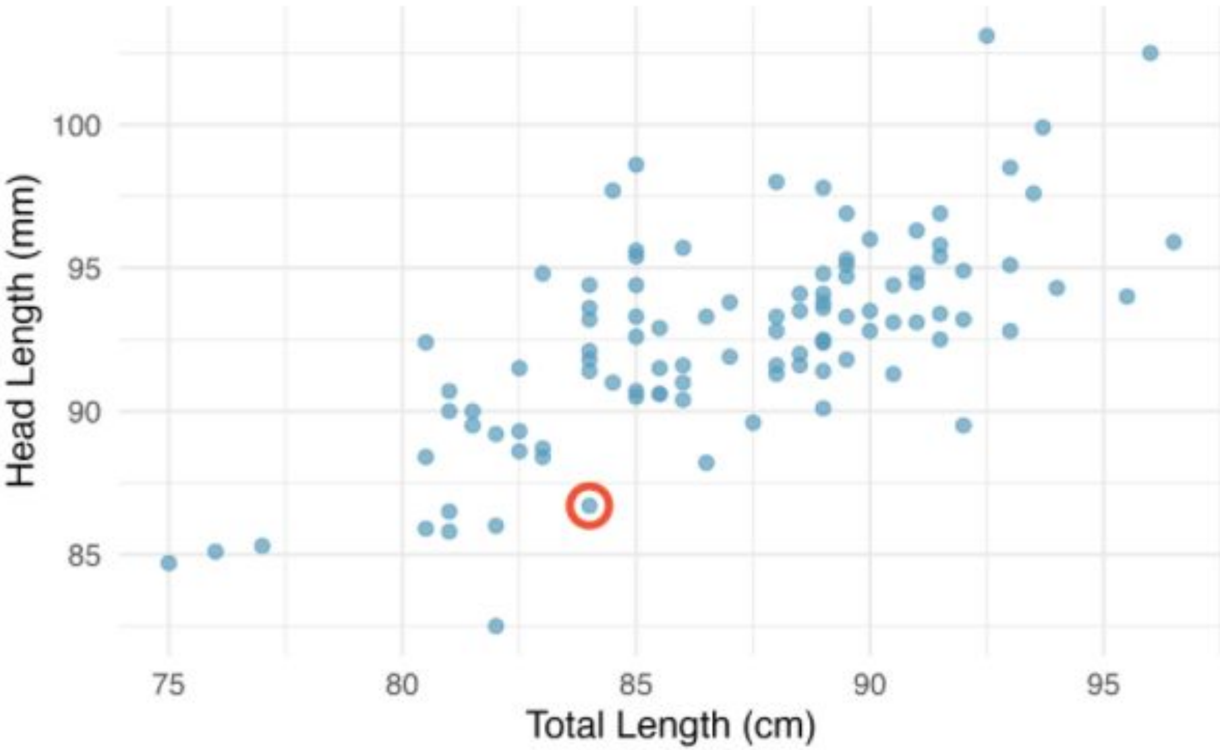
(1) Strong, negative, linear association
(2) Moderately strong, positive, linear association
(3) Weak, negative association

When not to use linear modeling!



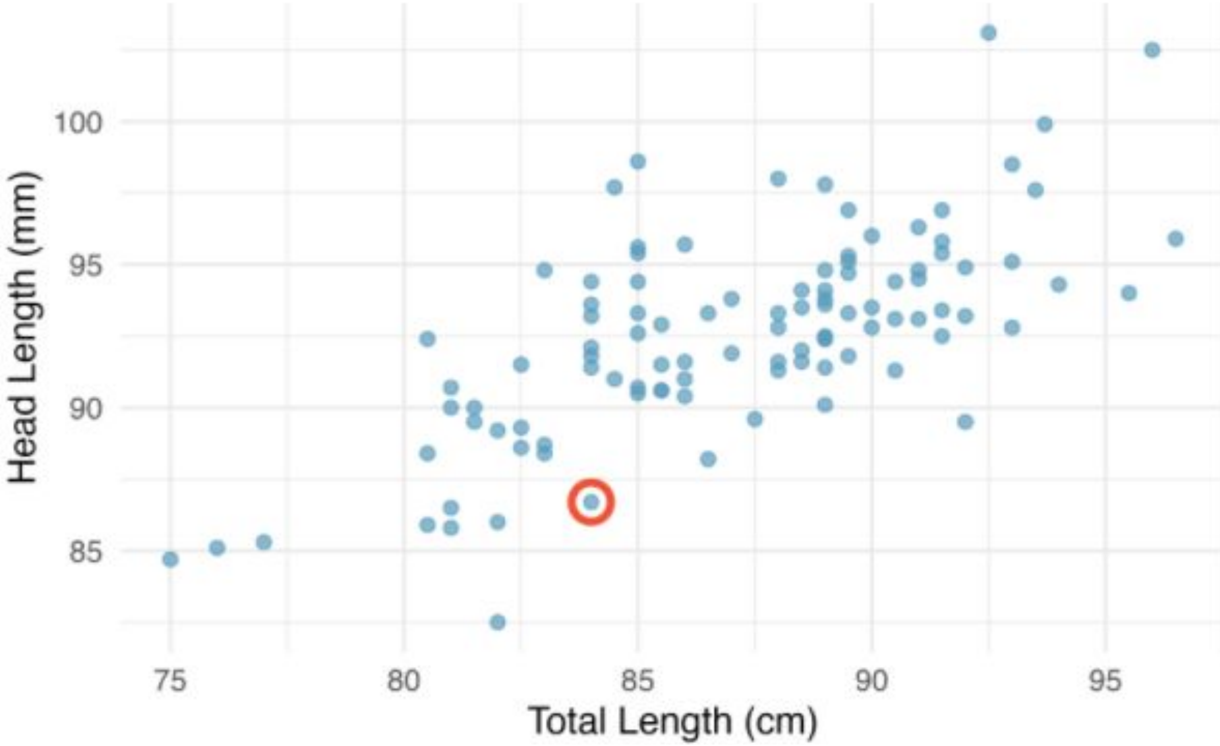
Fitting a linear model is not useful in this situation.

Total length and head length of the Brushtail Possum



Brushtail possums are marsupials that live in Australia. Researchers captured 104 of these animals and took body measurements before releasing the animals back into the wild.

Total length and head length of the Brushtail Possum



Shape: Linear

Strength: Moderately strong

Direction: Positive

Explanatory or Predictor Variable: Total Length (cm)

Response Variable: Head Length (mm)

Describing scatterplots

Address:

Shape: Linear

Strength: Moderately strong

Direction: Positive

Use:

G - description of the **G**eneral trend

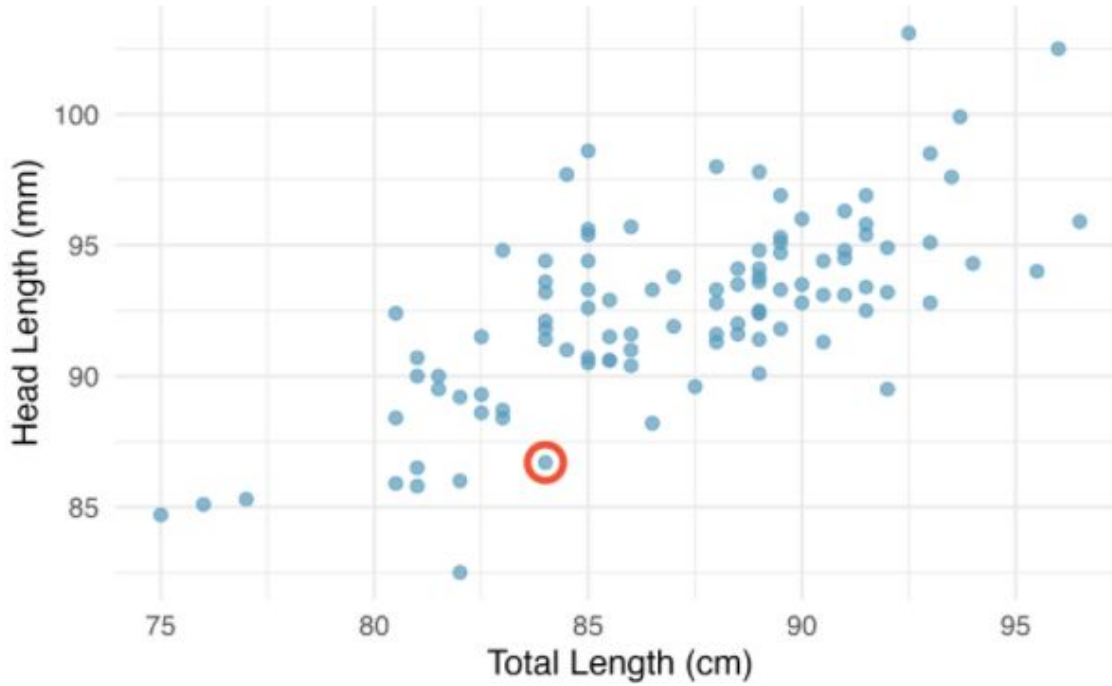
E - give illustrative **E**xamples of whatever point you make

E - list some examples of **E**xceptions to the general trend

As usual: Remember to always speak in context!

Total length and head length of the Brushtail Possum

This scatterplot shows the relationship between the total length of the brushtail possum and its head length. Generally, there is a positive linear association between these variables: possums with shorter bodies also tend to



have smaller heads, and possums with longer bodies tend to have larger heads. For example, one possum is about 81cm long in total, and has an 86mm head length, and there is another that is about 93cm in total length and has a head length of 100mm. The relationship is only moderately strong, meaning that there is some variation in head length for given values of total length. For example, for a total body length measurement of 85cm, the head length varies from about 91-98mm. [There are some exceptions to the general trend. For example, …]